

NAME :Jacques

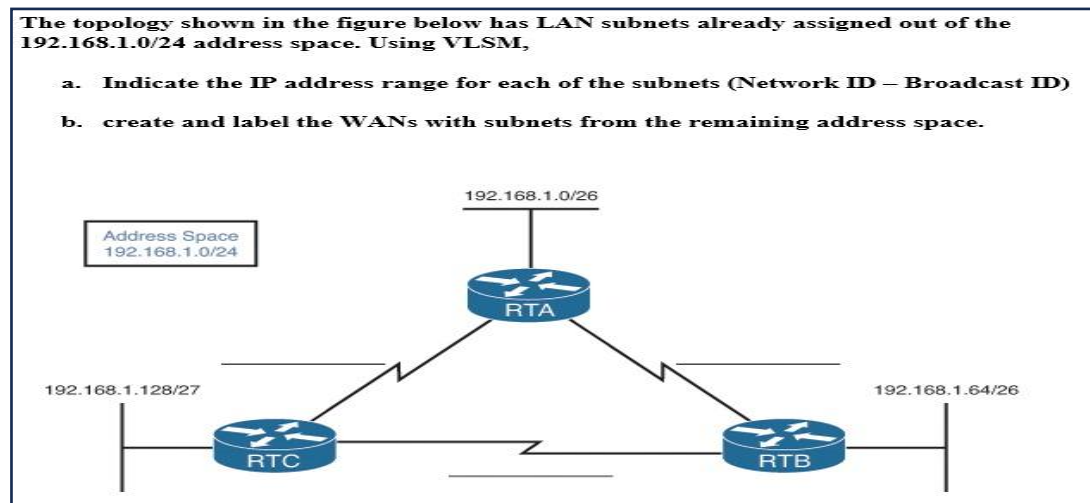
Monday, October 27th 2025

LASTNAME : RENANAGA

INDEX N : BSc IS/2024/02/0029

LEVEL :200.2

ASSIGNMENT TO BE SUBMITTED



Address space: 192.168.1.0/24

This means all IPs from 192.168.10 to 192.168.1.255 are available.

From the diagram:

RTA LAN subnet: 192.168.1.0/26

Subnet mask: 255.255.255.192 from 11111111.11111111.11111111.00000000

Number of Network: $2^2 = 4$ Network

Number of Host address: $2^n - 2$ from $2^6 - 2 = 64 - 2 = 62$

64 block size

62 host address

Network ID: 192.168.1.0

First host address: 192.168.1.1

Last host address: 192.168.1.62

Broadcast: 192.168.1.63

RTB LAN subnet: 192.168.1.64/26

Subnet mask: 255.255.255.192 from 11111111.11111111.11111111.00000000

Number of Network: $2^2 = 4$ Network

Number of Host address: $2^n - 2$ from $2^6 - 2 = 64 - 2 = 62$

64 block size

62 host address

Network ID: 192.168.1.64

First host address: 192.168.1.65

Last host address: 192.168.1.126

Broadcast: 192.168.1.127

RTC LAN subnet: 192.168.1.128/27

Subnet mask: 255.255.255.224 from 11111111.11111111.11111111.00000000

Number of Network: $2^3 = 8$ Network

Number of Host address: $2^n - 2$ from $2^5 - 2 = 32 - 2 = 30$

32 block size

30 host address

Network ID: 192.168.1.128

First host address: 192.168.1.129

Last host address: 192.168.1.158

Broadcast: 192.168.1.159

- b. Create and label the WANs with subnets from the remaining address space.

After 192.168.1.159, the remaining range is: 192.168.1.160-192.168.1.255

Each WAN link only needs 2 usable IPs, so we use /30 subnets(4 address total per subnet).

WAN RTA-RTB subnet: 192.168.1.160/30

Subnet mask: 255.255.255.252

Number of Network: $2^2 = 4$ *Network*

Number of Host address: $32 - 30 = 2$ *host address*

4 block size

2 host address

Network ID: 192.168.1.160/30

First host address: 192.168.1.161

Last host address: 192.168.1.162

Broadcast: 192.168.1.163

WAN RTA-RTC subnet: 192.168.1.164/30

Network ID: 192.168.1.164/30

First host address: 192.168.1.165

Last host address: 192.168.1.166

Broadcast: 192.168.1.167

WAN RTA-RTC subnet: 192.168.1.168/30

Network ID: 192.168.1.168/30

First host address: 192.168.1.169

Last host address: 192.168.1.170

Broadcast: 192.168.1.171

